

CA4 Liver

Imports

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
from numpy import interp

from sklearn.model_selection import train_test_split,
cross_val_predict, GridSearchCV, StratifiedKFold
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.decomposition import PCA
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
as LDA
from sklearn.metrics import f1_score, ConfusionMatrixDisplay,
roc_curve, auc
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression

C = [0.0001, 0.001, 0.01, 0.1, 1.0, 10.0, 100.0, 1000.0]
n_components = [1, 2, 3, 4, 5, 6]
gamma = [0.0001, 0.001, 0.01, 0.1, 1.0, 10.0, 100.0]
```

Reading data

```
# Reading the train and test data
df_train = pd.read_csv('data/train.csv', index_col=[0])
df_test = pd.read_csv('data/test.csv', index_col=[0])

# Removing unnecessary index column from train and test data
df_train.drop('index', axis=1, inplace=True)
df_test.drop('index', axis=1, inplace=True)
```

Data exploration and visualisation

```
# Finding out what types of data are in the dataset
df_train.dtypes

# Since there are categorical data, I have to encode these to
numerical values.
```

AFP (ng/mL)	float64
ALP (U/L)	float64
ALT (U/L)	float64

AST (U/L)	float64
Age	int64
Albumin (g/dL)	float64
Alcohol_Use (yes/no)	object
Bilirubin (mg/dL)	float64
CRP (mg/L)	float64
Diabetes (yes/no)	object
Fibroscan (kPa)	float64
GGT (U/L)	float64
Gender	object
Hemoglobin (g/dL)	float64
IL-6 (pg/mL)	float64
Obesity (yes/no)	object
PT/INR	float64
Platelets (10 ⁹ /L)	float64
RBC (10 ¹² /L)	float64
Serum_Ammonia (μmol/L)	float64
Serum_Copper (μg/dL)	float64
Serum_Creatinine (mg/dL)	float64
Serum_Iron (μg/dL)	float64
Serum_Lactate (mmol/L)	float64
Serum_Urea (mg/dL)	float64
Serum_Zinc (μg/dL)	float64
TIBC (μg/dL)	float64
Transferrin_Saturation (%)	float64
WBC (10 ⁹ /L)	float64
pH	float64
Diagnosis	object
dtype:	object

Checking if there are any missing values/NaN

```
df_train.isna().sum()
```

AFP (ng/mL)	0
ALP (U/L)	0
ALT (U/L)	0
AST (U/L)	0
Age	0
Albumin (g/dL)	0
Alcohol_Use (yes/no)	0
Bilirubin (mg/dL)	0
CRP (mg/L)	0
Diabetes (yes/no)	0
Fibroscan (kPa)	0
GGT (U/L)	0
Gender	0
Hemoglobin (g/dL)	0
IL-6 (pg/mL)	0
Obesity (yes/no)	0
PT/INR	0

Platelets (10 ⁹ /L)	0
RBC (10 ¹² /L)	0
Serum_Ammonia (μmol/L)	0
Serum_Copper (μg/dL)	0
Serum_Creatinine (mg/dL)	0
Serum_Iron (μg/dL)	0
Serum_Lactate (mmol/L)	0
Serum_Urea (mg/dL)	0
Serum_Zinc (μg/dL)	0
TIBC (μg/dL)	0
Transferrin_Saturation (%)	0
WBC (10 ⁹ /L)	0
pH	0
Diagnosis	0

dtype: int64

There are no missing values/NaN's, and therefore I do not need to impute.

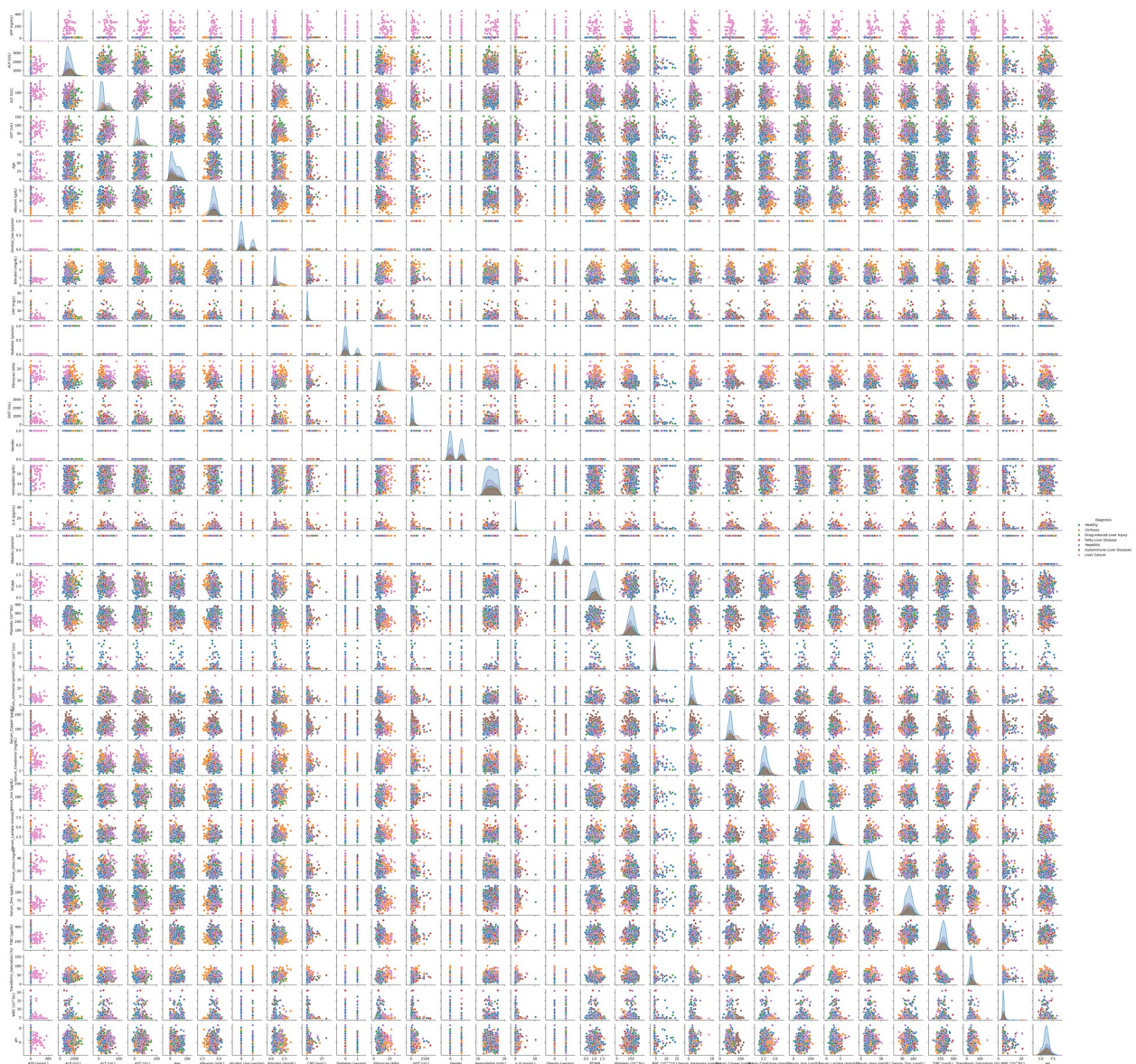
Data cleaning

```
# Sorting out columns with categorical data
cat_cols = df_train.select_dtypes(include=['object']).columns[:-1]

# for-loop that loops through columns in cat_cols and encode the
# categorcal data to zeros and ones.
for i in cat_cols:
    class_mapping = {label: idx for idx, label in
enumerate(np.unique(df_train[i]))}
    df_train[i] = df_train[i].map(class_mapping)

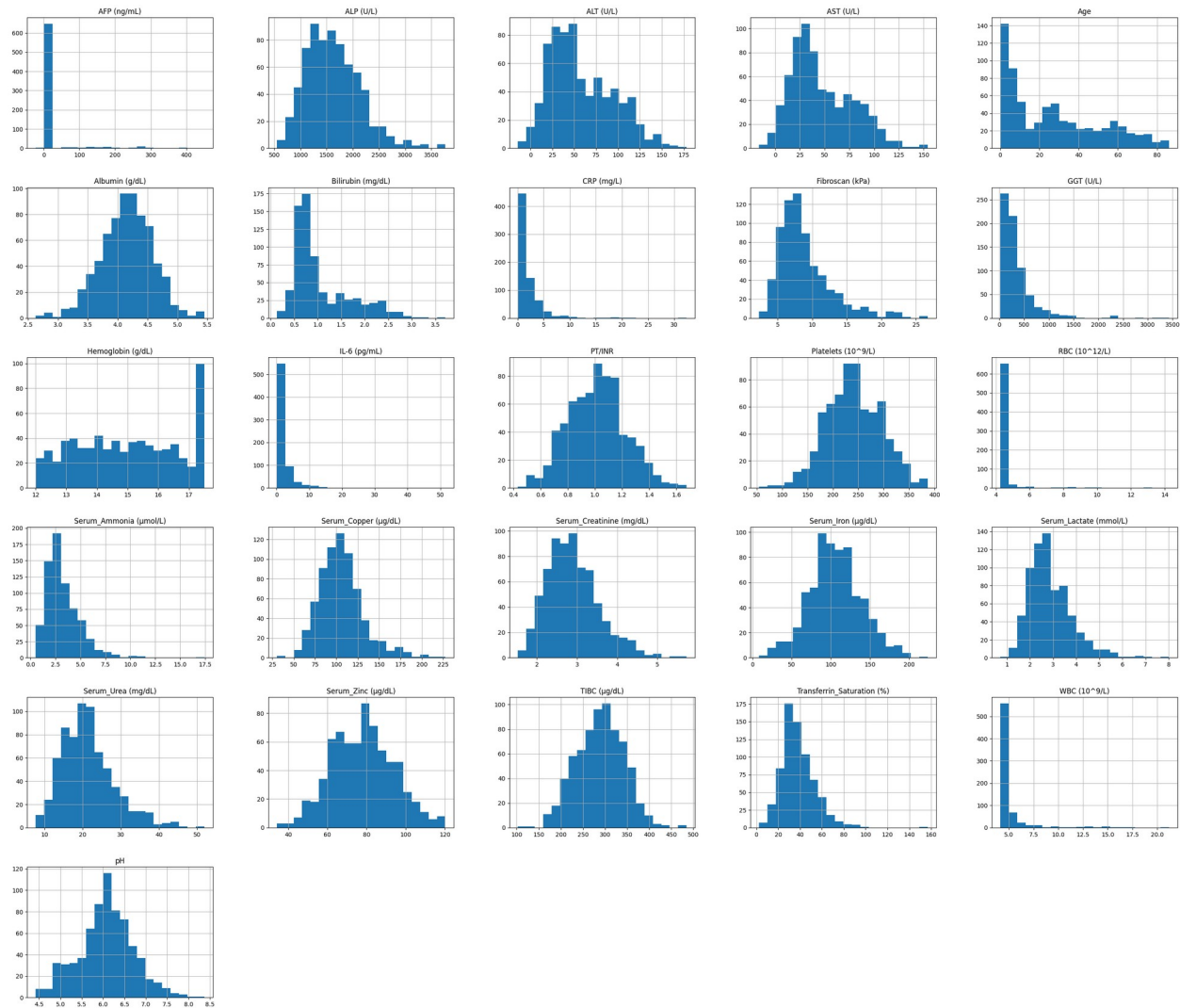
# A pairplot to visualize relationship between features and the
# target.
sns.pairplot(df_train, hue='Diagnosis', height=1.5)

<seaborn.axisgrid.PairGrid at 0x2d5d1fd90>
```



The pairplot gives an overview of the relationship between the features, but it is difficult to get meaningful information from since it so small. However it visualizes to a certain degree that there are some outliers accross features. It also visualize that target labels tend to gather as clusters, for example 'Chiriosis' (orange) and 'Liver cancer' (pink).

```
# Histogram of the features in the dataset
df_train.hist(bins=20, figsize=(35, 30))
plt.show()
```



The histogramplot indicates outliers in the dataset, for example in the features 'AFP (ng/mL)' and 'WBC (10⁹/L)' we see values that are significantly distant from the rest of the data. The distribution of the 'Age' feature makes sense, since the data most likely is collected from patients in all age groups. For the 'Hemoglobin (g/dL)' I do not have an explanation of the distribution.

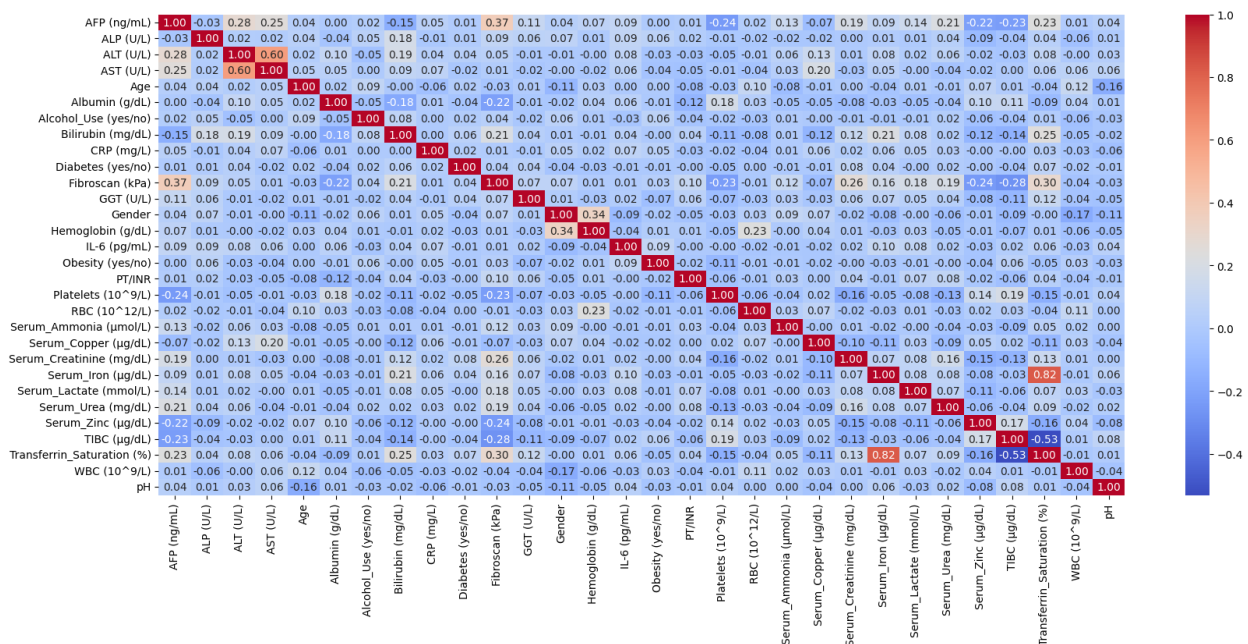
Outliers

With my comments about the pairplot and the histogramplot in mind, I do not want to remove any outliers from the data. That is, I believe removing the outliers will remove the real representation of the data, and also remove patterns in the data. Doing so will make my model not generalizing well on unseen data. Keeping the few outliers may also make my model more robust to outliers in datasets it has not been trained on.

```
num_cols = df_train.select_dtypes(include=['int64', 'float64'])
plt.figure(figsize=(20, 8))
```



```
heatmap = sns.heatmap(data=num_cols.corr(), annot=True, fmt=".2f",
cmap='coolwarm')
```



Most of the features shows little correlation to each other, but some are correlated significantly more than the rest. This will be handled by using dimension reduction during the preprocessing. Because there are so few features that are highly correlated to each other, I decide to use all the features for the modeling.

Data preprocessing and visualisation

```
# Defining the features and targets
X = df_train.drop('Diagnosis', axis=1)
y = df_train['Diagnosis']

# Splitting data in train and test.
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.3, random_state=42, stratify=y)

# Sorting between numerical and categorical columns.
cat_cols = ['Alcohol_Use (yes/no)', 'Diabetes (yes/no)', 'Gender',
'Obesity (yes/no)']
num_cols = X_train.columns.drop(cat_cols).to_list()

# Only scaling and applying PCA to the numerical data.
preprocessor = ColumnTransformer(
    transformers=[
        ('num', Pipeline([('scaler', StandardScaler()), ('pca',
PCA())]), num_cols),
        ('cat', 'passthrough', cat_cols)
    ])
])
```

```

# Making a pipeline for the different dimension reduction methods.
pca_pipeline = Pipeline([('scaler', preprocessor)
                          ])

lda_pipeline = Pipeline([('scaler', preprocessor),
                          ('LDA', LDA())])

```

Modelling

```

# Several nested pipelines to find out which one performs the best.
lr_pcapipeline = Pipeline([
    ('pca_pipeline', pca_pipeline),
    ('lr', LogisticRegression(penalty='l2', random_state=42))
])

lr_ldapipeline = Pipeline([
    ('lda_pipeline', lda_pipeline),
    ('lr', LogisticRegression(penalty='l2', random_state=42))
])

svm_pcapipeline = Pipeline([
    ('pca_pipeline', pca_pipeline),
    ('svm', SVC(kernel='linear', random_state=42))
])

svm_ldapipeline = Pipeline([
    ('lda_pipeline', lda_pipeline),
    ('svm', SVC(kernel='linear', random_state=42))
])

# Cross validation prediction of the data for each of the models.
pca_y_hat_lr = cross_val_predict(lr_pcapipeline, X_train, y_train,
cv=10)
lda_y_hat_lr = cross_val_predict(lr_ldapipeline, X_train, y_train,
cv=10)
pca_y_hat_svm = cross_val_predict(svm_pcapipeline, X_train, y_train,
cv=10)
lda_y_hat_svm = cross_val_predict(svm_ldapipeline, X_train, y_train,
cv=10)

# Computing the f1-score for the predictions made by different models.
f1_pca_lr = f1_score(y_train, pca_y_hat_lr, average='macro')
f1_lda_lr = f1_score(y_train, lda_y_hat_lr, average='macro')
f1_pca_svm = f1_score(y_train, pca_y_hat_svm, average='macro')
f1_lda_svm = f1_score(y_train, lda_y_hat_svm, average='macro')

# Printing the f1-score from the different models.
print(f'Logistic Regression with PCA gives F1-score: {round(f1_pca_lr

```

```

* 100, 2)}}')
print(f'Logistic Regression with LDA gives F1-score: {round(f1_lda_lr
* 100, 2)}}')
print(f'Support Vector Machine with PCA gives F1-score:
{round(f1_pca_svm * 100, 2)}}')
print(f'Support Vector Machine with LDA gives F1-score:
{round(f1_lda_svm * 100, 2)}}')

```

```

Logistic Regression with PCA gives F1-score: 70.76
Logistic Regression with LDA gives F1-score: 74.25
Support Vector Machine with PCA gives F1-score: 67.48
Support Vector Machine with LDA gives F1-score: 72.67

```

Hyperparameter tuning of the best performing logistic regression pipeline.

A logistic Regression pipeline with no

```

logreg_pipeline = Pipeline([
    ('lda', lda_pipeline),
    ('logreg', LogisticRegression())
])

```

The parameter grid for the logistic regression pipeline. Made to a list of dictionaries.

This is to ensure the grid search find the best possible combination of hyperparameters.

```

logreg_param_grid = [
    {'lda__LDA__n_components' : n_components, 'logreg__penalty' :
['l1'], 'logreg__solver' : ['liblinear'], 'logreg__C' : C},
    {'lda__LDA__n_components' : n_components, 'logreg__penalty' :
['l2'], 'logreg__solver' : ['liblinear', 'newton-cg'], 'logreg__C' :
C}
]

```

Splitting the dataset into 5-folds for the grid search.

```

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

```

Grid search over a 5-fold cross-validation defined by cv.

```

logreg_gridsearch = GridSearchCV(estimator=logreg_pipeline,
param_grid=logreg_param_grid, scoring='f1_macro', cv=cv, verbose=1)

```

Fit the grid search to the training data.

```

logreg_gridsearch.fit(X_train, y_train)

```

Printing out the best parameters and the respective f1-score.

```

print('Best parameters: ', logreg_gridsearch.best_params_)
print('Best score: ', logreg_gridsearch.best_score_)

```

```

Fitting 5 folds for each of 144 candidates, totalling 720 fits
Best parameters: {'lda__LDA__n_components': 6, 'logreg__C': 10.0,

```



```

'logreg__penalty': 'l2', 'logreg__solver': 'newton-cg'}
Best score: 0.7564100086486971

# Hyperparametertuning of the best performing SVM pipeline.

# A SVM pipeline with no
svm_pipeline = Pipeline([
    ('lda', lda_pipeline),
    ('svm', SVC())
])

# The parameter grid for the SVM pipeline. Made to a list of
dictionaries.
# This is to ensure the grid search find the best possible combination
of hyperparameters.
svm_param_grid = [
    {'lda__LDA__n_components': n_components, 'svm__C' : C,
    'svm__kernel' : ['linear']},
    {'lda__LDA__n_components': n_components, 'svm__C' : C,
    'svm__kernel' : ['rbf'], 'svm__gamma' : gamma}
]

# Splitting the dataset into 5 folds for the grid search
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

# Grid search over a 5-fold cross-cvalidation defined by cv.
svm_gridsearch = GridSearchCV(estimator=svm_pipeline,
param_grid=svm_param_grid, scoring='f1_macro', cv=cv, verbose=1)

# Fit the grid search to the training data.
svm_gridsearch.fit(X_train, y_train)

# Printing out the best parameters and the respective f1-score.
print('Best parameters: ', svm_gridsearch.best_params_)
print('Best score: ', svm_gridsearch.best_score_)

Fitting 5 folds for each of 384 candidates, totalling 1920 fits
Best parameters: {'lda__LDA__n_components': 5, 'svm__C': 1.0,
'svm__gamma': 0.1, 'svm__kernel': 'rbf'}
Best score: 0.7396580376784592

# Evaluation of the logistic regression model after tuning the
hyperpareters.
best_logreg = logreg_gridsearch.best_estimator_
y_pred_logreg = best_logreg.predict(X_test)
f1_logreg = f1_score(y_test, y_pred_logreg, average='macro')
print(f'F1-score logreg after tuning hyperparameters: {f1_logreg}')

# Evaluation of the SVM model after tuning the hyperparameters.
best_svm = svm_gridsearch.best_estimator_
y_pred_svm = best_svm.predict(X_test)

```

```
f1_svm = f1_score(y_test, y_pred_svm, average='macro')
print(f'F1-score svm after tuning hyperparameters: {f1_svm}')
```

```
F1-score logreg after tuning hyperparameters: 0.7579350518048192
F1-score svm after tuning hyperparameters: 0.7400543575057462
```

The logistic regression model is best performing after tuning the hyperparameters, and im therefore using it for the kaggle submission.

Final evaluation

```
# Sorting out the columns in the test data with categorical data.
cat_cols_test = df_test.select_dtypes(include=['object']).columns

# for-loop that loops through cols in cat_cols_test and encode the
# categorcal data to zeros and ones.
for i in cat_cols_test:
    class_mapping = {label: i for i, label in
enumerate(np.unique(df_test[i]))}
    df_test[i] = df_test[i].map(class_mapping)

# Fit the model I am using for the Kaggle submission on the whole
# training dataset.
best_logreg.fit(X, y)

# Prediction on the test dataset.
y_pred = best_logreg.predict(df_test)
```

Confusion matrix of best model

```
# Splitting the trainig data 60/40 as specified in the CA description.
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.4, random_state=42, stratify=y)

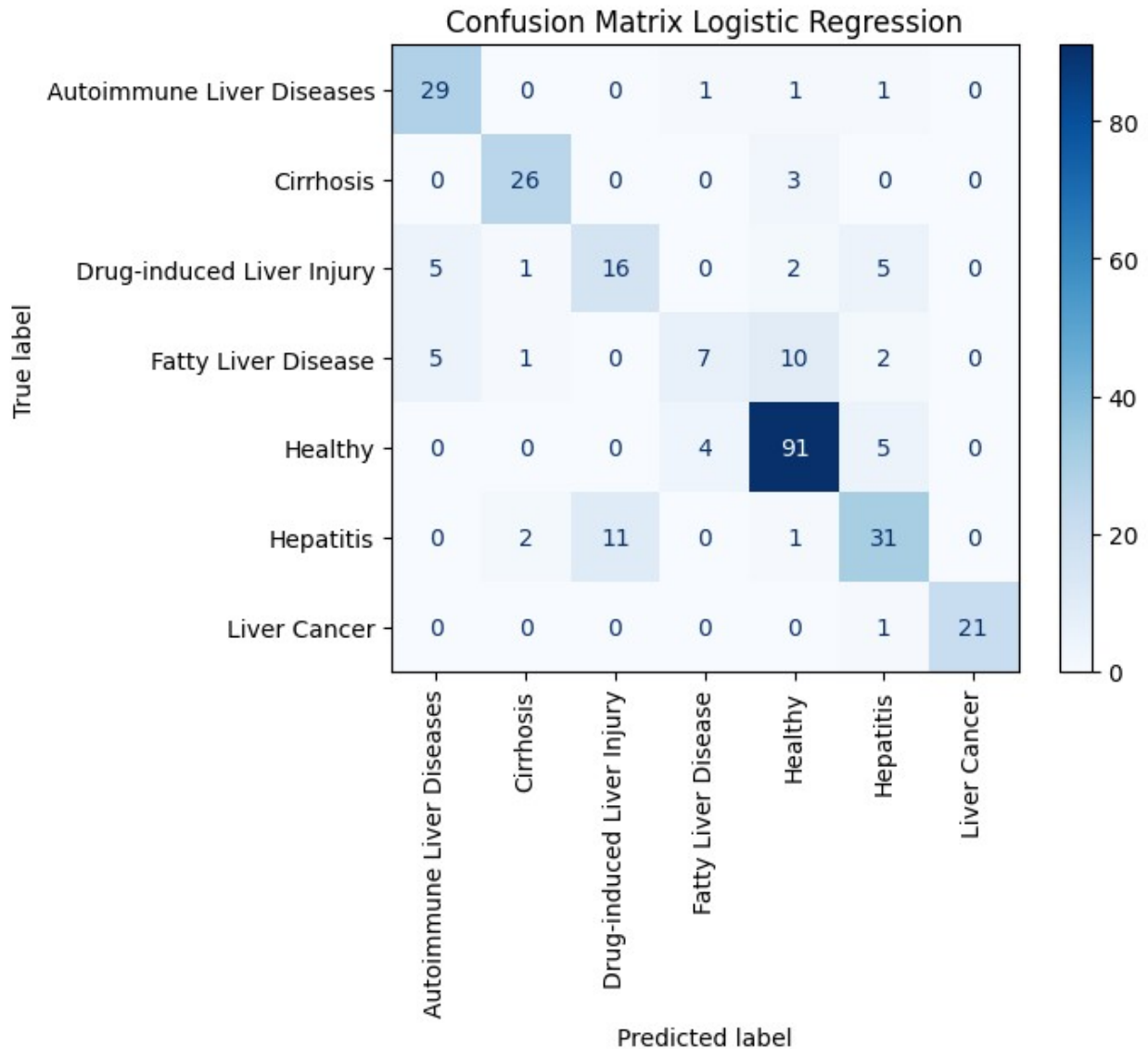
# Fit the best model to the new split of training dataset.
best_logreg.fit(X_train, y_train)

# Predicton on the new split.
y_pred_conf = best_logreg.predict(X_test)

# Creating and plotting the confusion matrix
conf_mat = ConfusionMatrixDisplay.from_predictions(y_test,
y_pred_conf, cmap=plt.cm.Blues,)
plt.title('Confusion Matrix Logistic Regression')
plt.xticks(rotation=90)

(array([0, 1, 2, 3, 4, 5, 6]),
 [Text(0, 0, 'Autoimmune Liver Diseases'),
  Text(1, 0, 'Cirrhosis'),
  Text(2, 0, 'Drug-induced Liver Injury'),
```

```
Text(3, 0, 'Fatty Liver Disease'),
Text(4, 0, 'Healthy'),
Text(5, 0, 'Hepatitis'),
Text(6, 0, 'Liver Cancer')]]
```



Confusion matrix

The confusion matrix indicates that my model performs well. That is, the confusion matrix is diagonal dominant. This means that most label predictions were true for the respective classes. However, the model performs poorly when predicting 'Fatty liver diseases'. It predicts 'Healthy' more times than the correct label 'Fatty liver disease'. The same goes for predicting 'Hepatitis', here the model predicts 'Drug-induced Liver Injury' almost 25% of the times.

Kaggle submission

```
# Creating Kaggle subbmission .csv file.
y_final = pd.DataFrame(y_pred, columns=["Diagnosis"])
y_final.index.name = "index"
y_final.to_csv("predictions3.csv")
```

Binary target variables

For this part I have used and modified the code script "ROC_AUC_with_LR.ipynb" from lecture 16, and used tips provided on the padlet.

```
# Defining the features to be used
X_binary = df_train.drop('Diagnosis', axis=1)

# Encoding target labels to the numerical values 0 ('Healthy') and 1 ('Diseased')
y_binary = np.array([0 if label == 'Healthy' else 1 for label in df_train['Diagnosis']])

# Splitting data in train and test.
X_train, X_test, y_train, y_test = train_test_split(X_binary, y_binary, test_size=0.3, random_state=42)

# Sorting between categorical and numerical columns
cat_cols = ['Alcohol_Use (yes/no)', 'Diabetes (yes/no)', 'Gender', 'Obesity (yes/no)']
num_cols = X_train.columns.drop(cat_cols).tolist()

# Preprocessor for the pipeline
preprocessor = ColumnTransformer([
    ('num', StandardScaler(), num_cols), # Applying StandardScaler to standardize features.
    ('cat', OneHotEncoder(), cat_cols) # Applying OneHotEncoder for categorical columns.
])

# Pipeline
pipe_binary = Pipeline([
    ('scaler', preprocessor),
    ('clf', LogisticRegression(penalty='l2', random_state=1, C=10.0, solver='newton-cg'))
])

# Setting up a 5-fold cross-validation
cv_ = list(StratifiedKFold(n_splits=5).split(X_train, y_train))

# Plotting
fig = plt.figure(figsize=(7, 5)) # Creates a figure
mean_tpr = 0.0 # Initializing the mean true
```

```

positive rate
mean_fpr = np.linspace(0, 1, 100)      #
all_tpr = []                          # Initializes a list to store
the tpr values for each fold in the cross-validation.

for i, (train, test) in enumerate(cv_): # for-loop that iterates over
the folds in the cross-val.
    probas = pipe_binary.fit(X_train.iloc[train],
y_train[train]).predict_proba(X_train.iloc[test]) # Fits the pipeline
on the training data of the current fold,

# before using the fitted model to predict class probabilities
# on the test data.

    fpr, tpr, thresholds = roc_curve(y_train[test], probas[:, 1],
pos_label=1) # Calculating true positive rate, false
positive rate and thresholds for ROC curve with

# true labels and predicted probabilities for the
    mean_tpr += interp(mean_fpr, fpr, tpr)
# Adding interpolated tpr values from current fold to

# to mean_tpr.
    roc_auc = auc(fpr, tpr)
# Calculating auc for current fold.
    plt.plot(fpr, tpr, label='ROC fold %d (area = %0.2f)' % (i+1,
roc_auc)) # Plotting ROC curve for current fold.

plt.plot([0, 1], [0, 1], linestyle='--', color=(0.6, 0.6, 0.6),
label='random guessing') # Plotting ROC curve of a random guess.

mean_tpr /= len(cv_) # Calculates mean true
positive rate
mean_tpr[0] = 0.0 # Initializing the mean ROC
curve start in origo (0, 0).
mean_tpr[-1] = 1.0 # Initializing the mean ROC
curve end in (1, 1).
mean_auc = auc(mean_fpr, mean_tpr) # Calculating AUC for the mean
ROC curve.
plt.plot(mean_fpr, mean_tpr, 'k--', label='Mean ROC (area = %0.2f)' %
mean_auc, lw=2) # Plotting mean ROC curve with AUC value as
label.

plt.plot([0, 0, 1], [0, 1, 1], linestyle=':', color='black',
label='perfect performance') # Creates the ROC curve for a perfect
performance classifier.

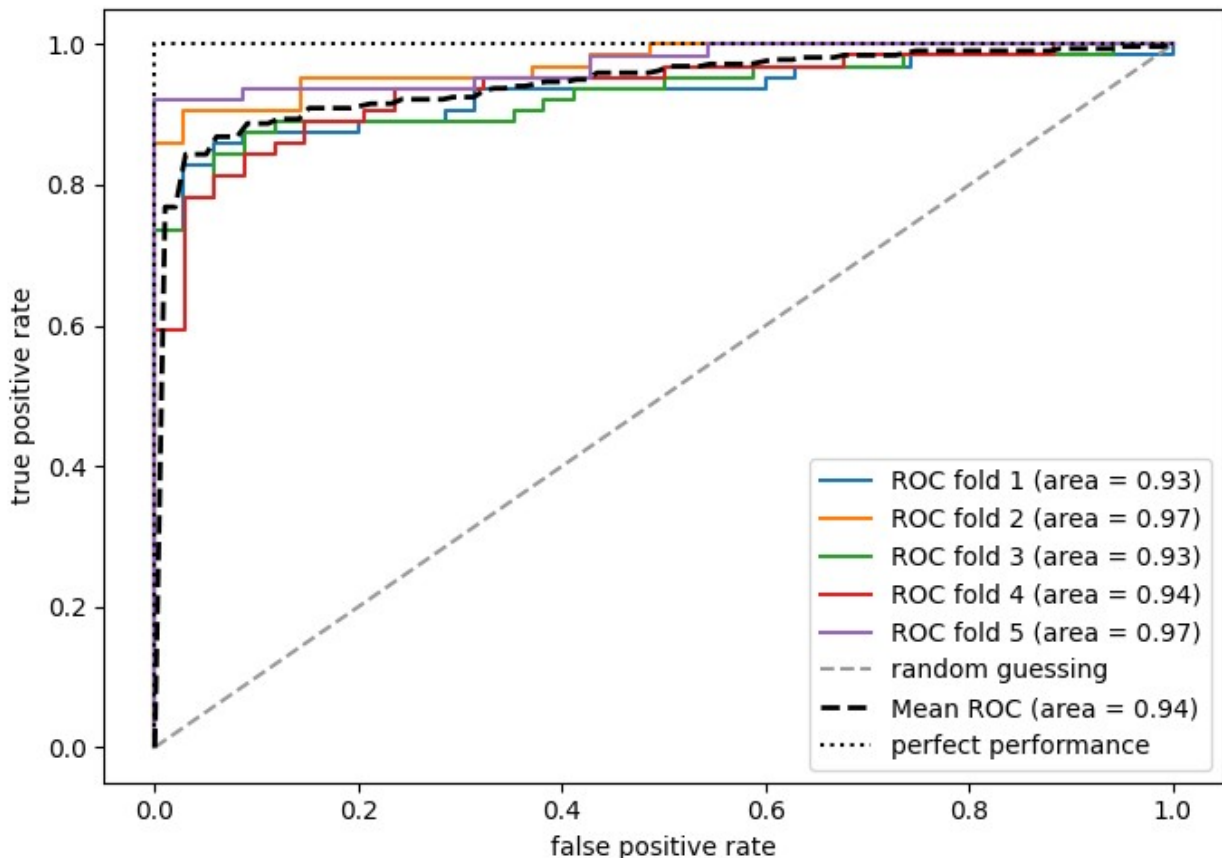
# xlim and ylim ensures that the whole ROC curve is visible in the
plot.

```



```
plt.xlim([-0.05, 1.05])
plt.ylim([-0.05, 1.05])

plt.xlabel('false positive rate') # Setting label for x-axis.
plt.ylabel('true positive rate') # Setting label for y-axis.
plt.legend(loc="lower right") # Placing the legend in the lower
                             # right corner of the figure.
plt.tight_layout() # Fitting the plot elements to the
                  # figure.
plt.show() # Displaying the ROC curve plot.
```



ROC-curve

The ROC-curves indicates that the model is able to discriminate between the classes well and is stable and consistent. That is, for each fold the ROC-curve is way above the random guessing line, the AUC scores for the folds range between 0.93 and 0.97, which is very well, and the AUC score for the mean ROC-curve is 0.94. It is also clearly to see that the model achieves a high true positive rate at a low false positive rate, which is very desirable. The ROC-curve indicates that the model will generalize well on unseen data.